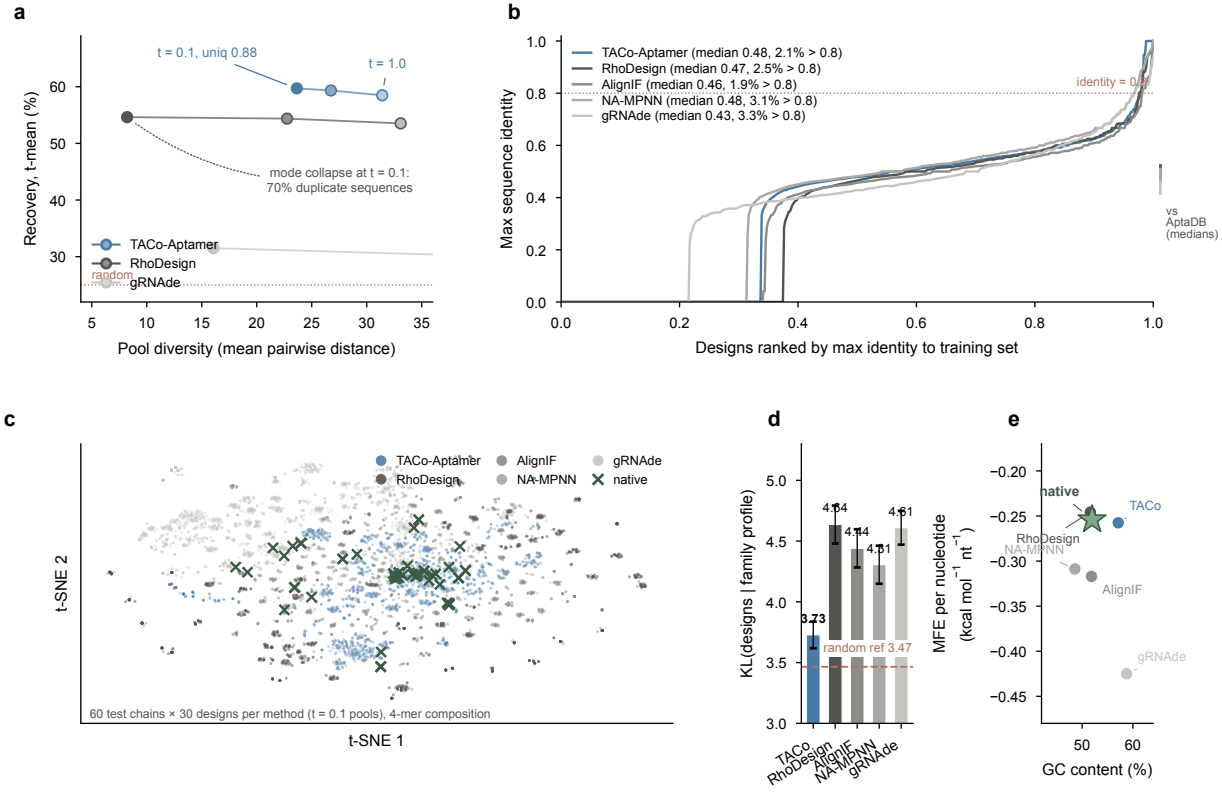

a
t = 0.1, uniq 0.88
t = 1.0
mode collapse at t = 0.1:
70% duplicate sequences
TACo-Aptamer
RhoDesign
gRNAde
random
Recovery, t-mean (%)
Pool diversity (mean pairwise distance)
b
TACo-Aptamer (median 0.48, 2.1% > 0.8)
RhoDesign (median 0.47, 2.5% > 0.8)
AlignIF (median 0.46, 1.9% > 0.8)
NA-MPNN (median 0.48, 3.1% > 0.8)
gRNAde (median 0.43, 3.3% > 0.8)
identity = 0.8
vs AptaDB (medians)
Max sequence identity
Designs ranked by max identity to training set
c
TACo-Aptamer
RhoDesign
AlignIF
NA-MPNN
gRNAde
native
60 test chains × 30 designs per method (t = 0.1 pools), 4-mer composition
t-SNE 1
t-SNE 2
d
KL(designs | family profile)
3.73
4.64
4.44
4.31
4.61
random ref 3.47
TACo
RhoDesign
AlignIF
NA-MPNN
gRNAde
e
native
TACo
RhoDesign
NA-MPNN
AlignIF
gRNAde
MFE per nucleotide (kcal mol⁻¹ nt⁻¹)
GC content (%)